

UNNI ADITYA MOHAN

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EDUCATION

Northeastern University, Boston Master of Science in Robotics Relevant Coursework: Robot Mechanics and Control, Sensing and Navigation, Reinforcement Learning	Sep 24 – Present GPA: 3.82
National Institute of Technology Rourkela, India Bachelor of Technology in Mechanical Engineering with Honors Relevant Coursework: Thermal Engineering, Fluid and Solid Mechanics, Mechatronics, CAD, CFD	Aug 17 – Jun 21 CGPA: 8.12 (/10)

SKILLS

Programming: Python, C++, MATLAB
Design and Simulation Tools: SolidWorks, AutoCAD, ANSYS Mechanical, ANSYS Fluent, Simulink, COMSOL, Arduino
Frameworks, Libraries and Development Tools: ROS2, ROS, OpenCV

WORK EXPERIENCE

Project Engineer Thermax Ltd, Pune, India	Sep 21-May 24
- Acted as a primary coordinator between engineering, procurement and manufacturing team, identifying critical path delays in design approval and implemented agile adjustments that speeded up project delivery by 2 weeks. - Developed and trained a machine learning model to predict optimal steam drum design parameters, which automated the CAD design process, that saved 20 man-hours and improved accuracy. - Saved \$20K per project by designing a reusable transportation frame in AutoCAD for pressure parts.	

PROJECTS

Decentralized Multi-Robot Transportation Using Force Feedback Control	Mar 25 - Present
- Designed a decentralized force-based controller for leader-follower wheeled robots to enable scalable, coordinated object manipulation, incorporating beam-type load cells calibrated to ± 0.5 N resolution for real-time force feedback at 100 Hz. - Created a MATLAB Simulink simulation to validate the control strategy using a two-robot system	
Thermal Control System of the International Space Station (ISS) using ROS2 (A Collaborative open-source project by Space Station OS in partnership with the Japan Aerospace Exploration Agency (JAXA) to simulate ISS subsystems with ROS2)	Dec 24 – Feb 25
- Developed a ROS2-based thermal regulation system using a client-server architecture (in C++) to stabilize ISS thermal parameters under dynamic orbital conditions. - Implemented PID control with sensor feedback in ROS2 (Gazebo) to ensure performance under solar and varied heat loads.	
Pick and Place Robot with Obstacle Avoidance Using PincherX100 (Vision-Guided) (Team of 2)	Oct 24 - Dec 24
- Developed custom inverse kinematics (IK) algorithms in MATLAB. - Implemented trajectory planning based on velocity control mode, achieving smooth and precise arm motion. - Designed a custom collision detection and avoidance algorithm by bounding obstacles with virtual boxes, checking waypoint intersections, and rerouting paths based on the manipulator's geometric limits.	
Mechanical Design Lead for Autonomous Under Water Vehicle (AUV) (Team of 6) (This AUV was developed for underwater competitions like ROBOSUB, SAUVC and SAVe.)	Jan 18 - Mar 21
- Achieved 42% drag reduction and enhanced stability by reducing lift force through mechanical redesign and CFD analysis in ANSYS Fluent, optimizing hydrodynamic performance. - Performed topology optimization in ANSYS Mechanical, reducing chassis weight by 20% while maintaining structural integrity at 15ft depth (1.8x safety factor) - Developed a piston-based ballast tank with a response time of <10sec for full buoyancy at 15ft, a torpedo shooting mechanisms with 3-5m precision for the competition and a 3-fingered gripper with a capacity of carrying 1kg underwater.	
Design of Effi-Cycle Chassis and Suspension System (Team of 4)	Sep 17 - Apr 20
(Effi-cycle is a human-electric hybrid power vehicle developed for the SAE India competition. Designed for two passengers, it is powered by solar panels, combining human pedaling with electric power for enhanced efficiency.) - Lead the design team to develop a competition-compliant 15kg chassis and suspension system using CAD. - Performed FEA (Static Structural & Explicit Dynamics) in ANSYS Workbench to evaluate structural integrity under operational loads, including front, rear, and side impact scenarios.	

ACHIEVEMENTS

- Winners in SAVe 2019, held in IIT Madras, India.
- Semi-Finalists in the ROBOSUB 2019 competition, held in San Deigo, US.